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# Neutrinos via Charm Decays in Astrophysical Sources

TALK IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE

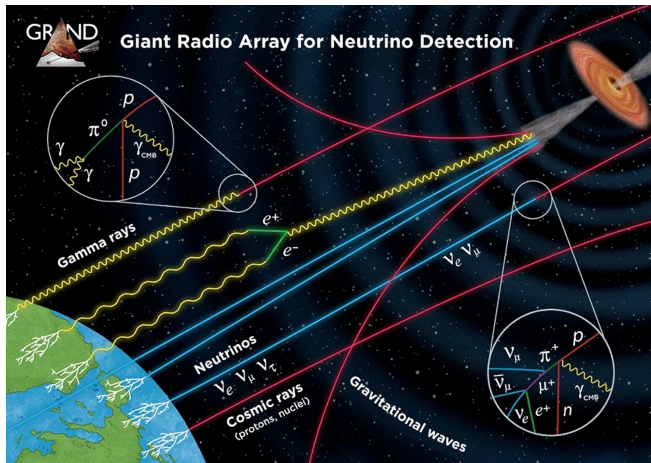
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Fritz Ali Agildere

**September 3, 2024**

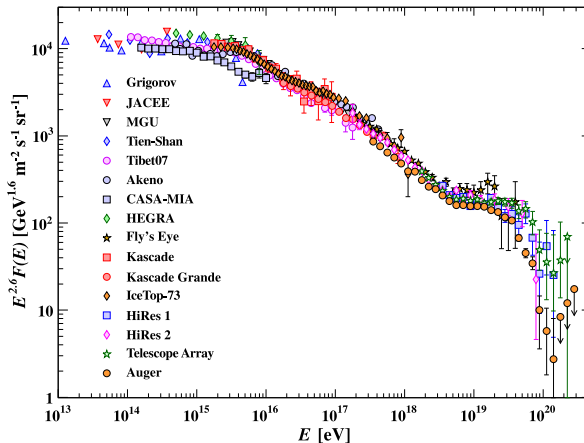
Technical University of Dortmund & Ruhr University Bochum

## Multimessenger Astrophysics



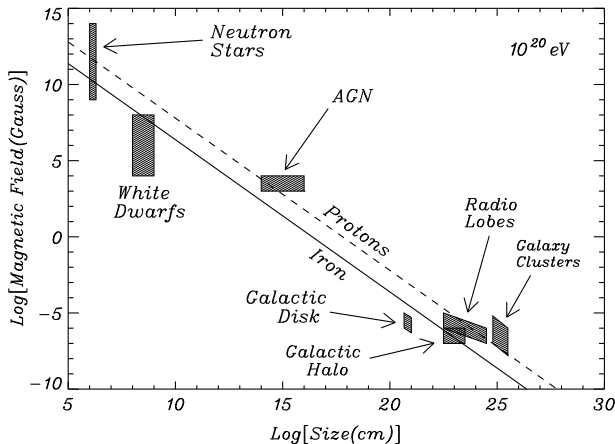
X. L. Qian et al.  
*Nucl. Sci. Tech.* (2021)

## Cosmic Ray Spectrum & Cutoff



K. A. Olive et al.  
*Chin. Phys. C* (2014)

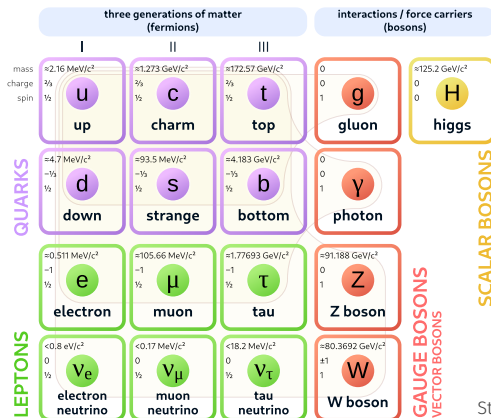
## Plausible Cosmic Ray Origins



A. V. Olinto  
AIP Conf. Proc. (2000)

## Elementary Particles & Hadrons

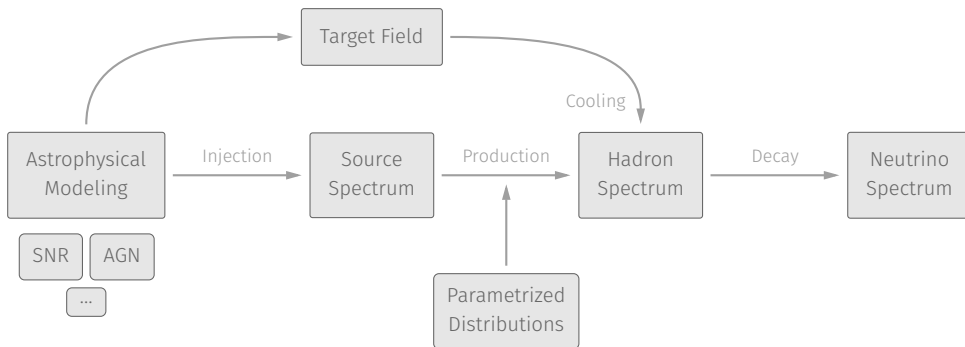
### Standard Model of Elementary Particles

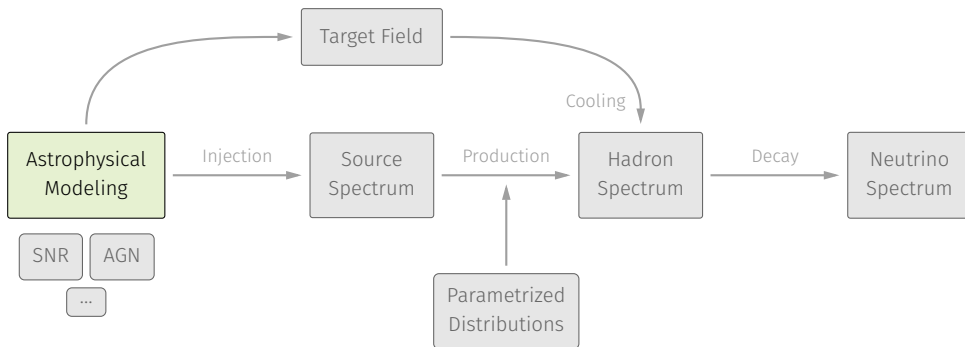


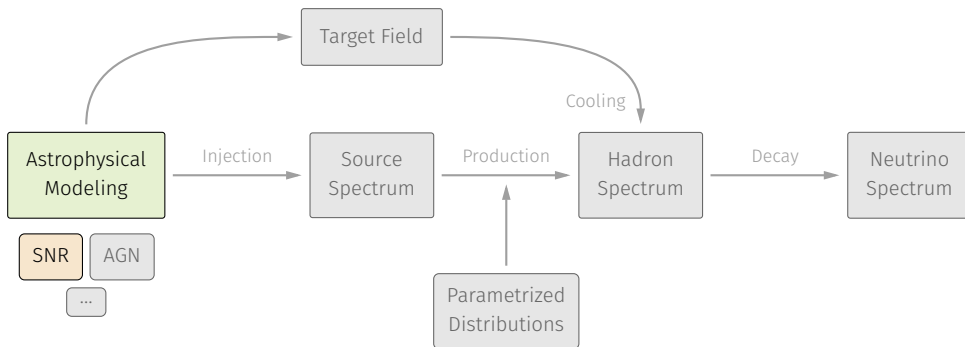
Standard Model  
Wikimedia Commons (2019)

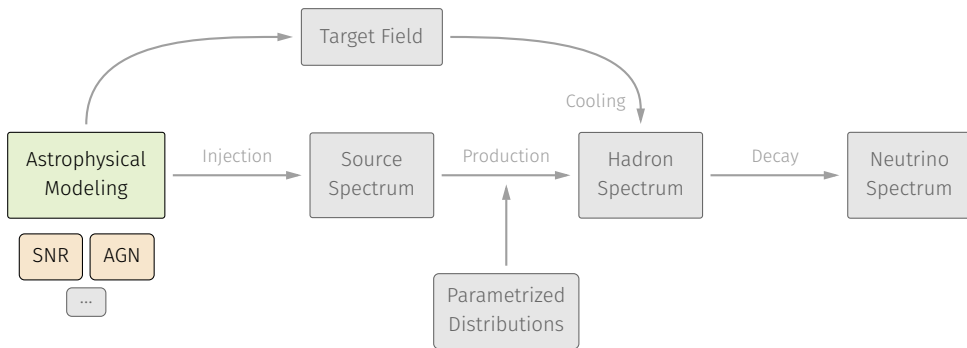
## Relevant Hadron Properties

| $h$           | $q_h$      | $\tau_h / \text{s}$   | $m_h / \text{GeV}$ | $s_h / \text{GeV}^2$ | $B_{h \rightarrow \nu}$ |
|---------------|------------|-----------------------|--------------------|----------------------|-------------------------|
| $p$           | $uud$      | $\infty$              | 0.94               | 15.98                | 0.000                   |
| $\pi^+$       | $u\bar{d}$ | $26.0 \times 10^{-9}$ | 0.14               | 10.23                | 1.000                   |
| $K^+$         | $u\bar{s}$ | $12.4 \times 10^{-9}$ | 0.49               | 12.62                | 0.636                   |
| $D_0$         | $c\bar{u}$ | $0.4 \times 10^{-12}$ | 1.86               | 0.45                 | 0.067                   |
| $D^+$         | $c\bar{d}$ | $1.0 \times 10^{-12}$ | 1.87               | 0.40                 | 0.176                   |
| $D_s^+$       | $c\bar{s}$ | $0.5 \times 10^{-12}$ | 1.97               | 0.71                 | 0.065                   |
| $\Lambda_c^+$ | $udc$      | $0.2 \times 10^{-12}$ | 2.29               | 1.61                 | 0.045                   |

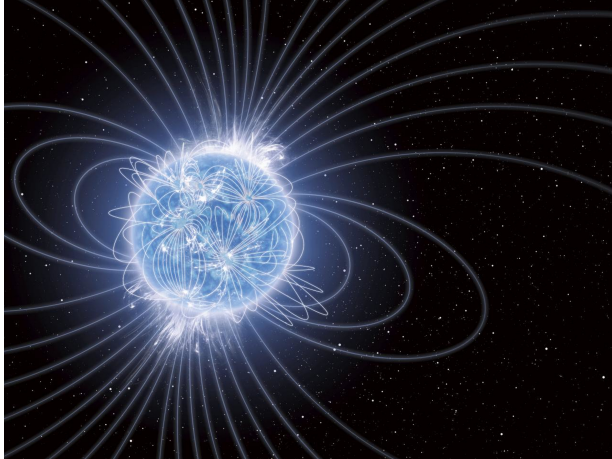






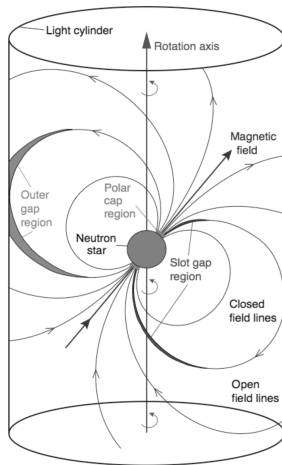


## Young Neutron Stars



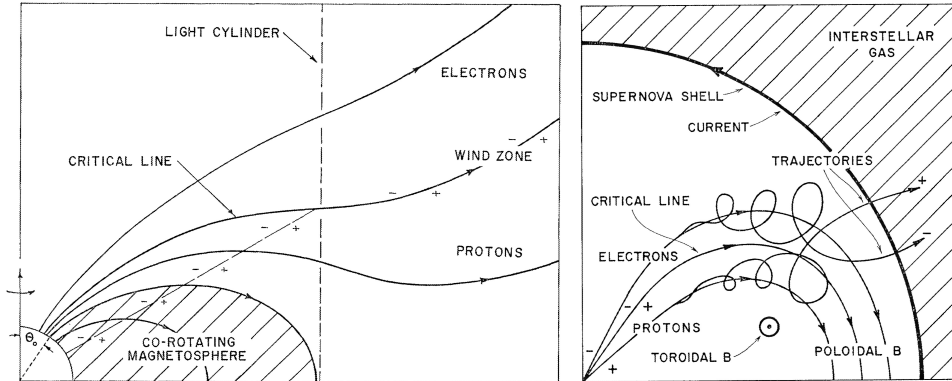
R. Takeshige  
*Magnetar* (2020)

## Dipolar Pulsar Model



E. Aliu et al.  
*Science* (2008)

## Particle Acceleration



P. Goldreich, W. H. Julian *ApJ* (1969)



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$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

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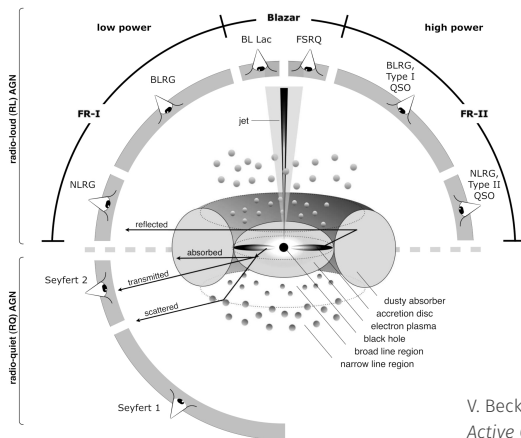
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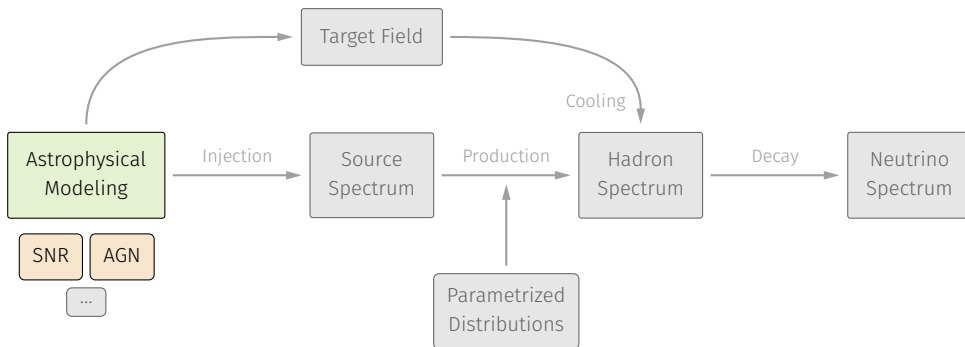
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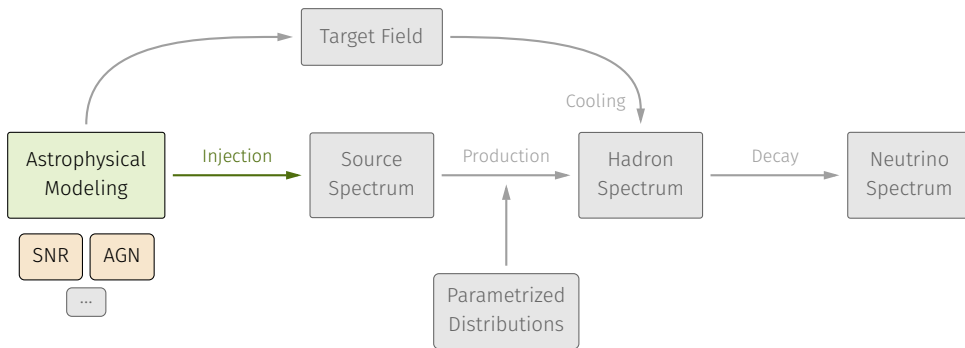
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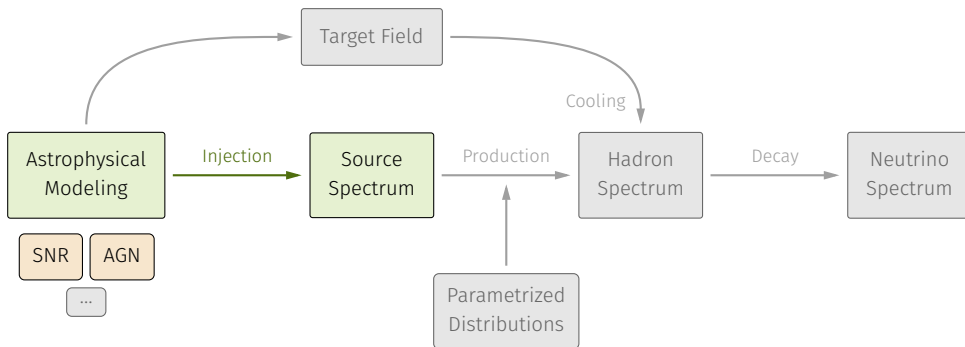
## Active Galactic Nuclei



V. Beckmann, C. Shrader  
*Active Galactic Nuclei* (2012)









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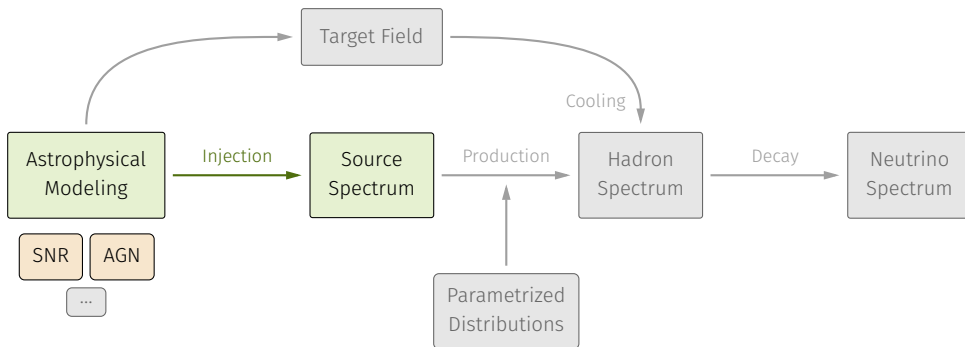
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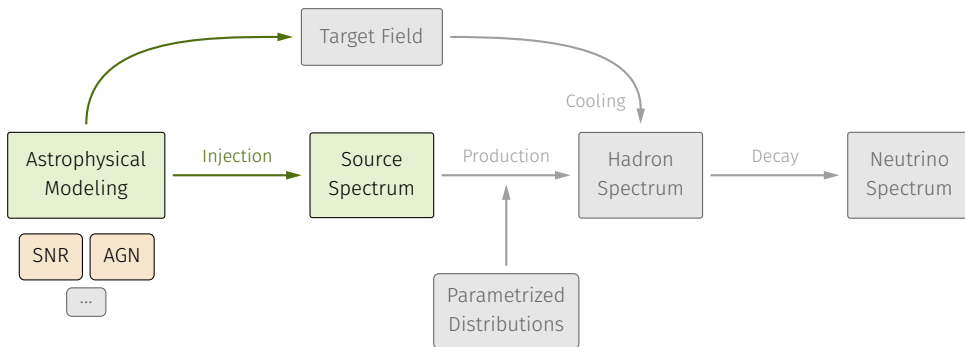
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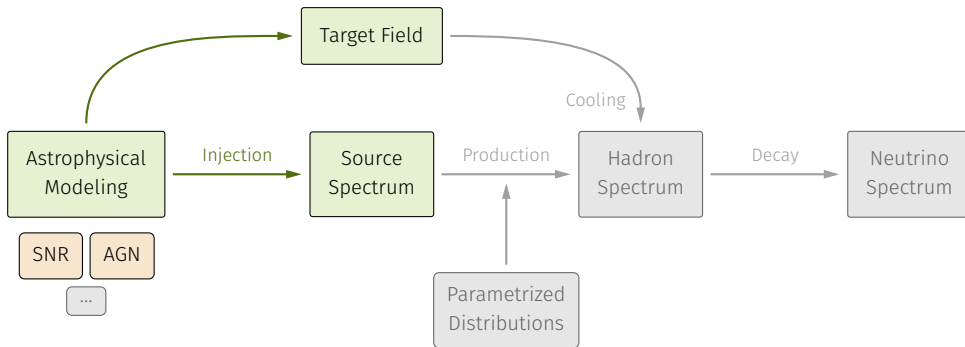
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$$\begin{array}{llll} I_{ns} = 10^{45} \text{ g cm}^2 & R_{ns} = 10^6 \text{ cm} & B_{ns} = 10^{14.5} \text{ G} & t_{sd} = 10^{3.5} \text{ s} \\ M_{ej} = 10 M_{\odot} & \beta_{ej} = 0.1 & \eta = 0.1 & \end{array}$$



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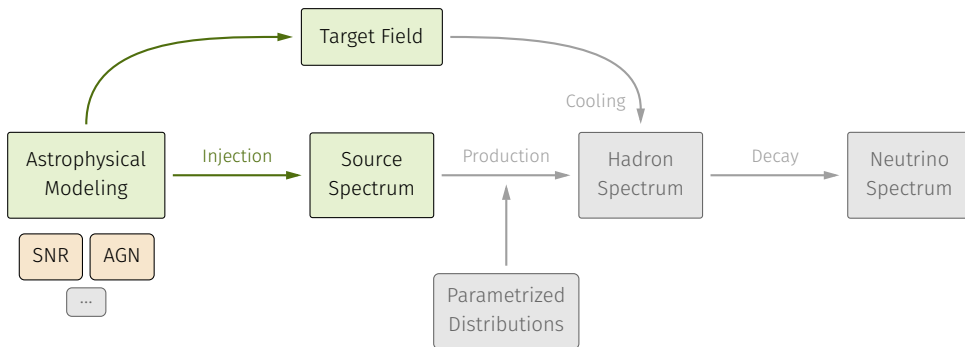
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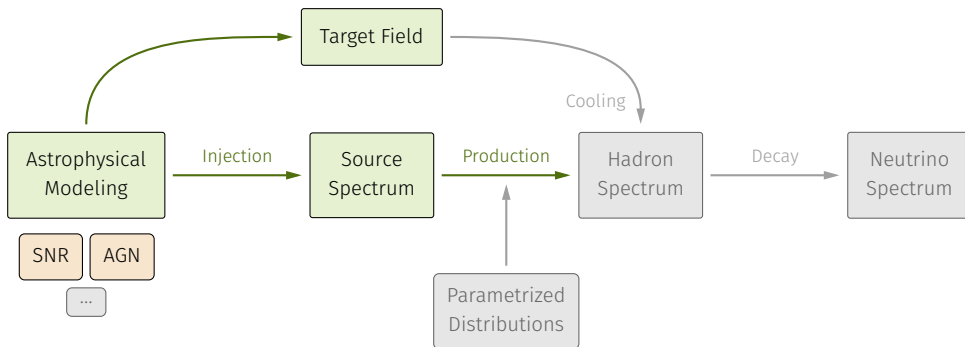
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$$n_{dc} = 10^{14} \text{ cm}^{-3} \qquad h = 10^{15} \text{ cm}$$







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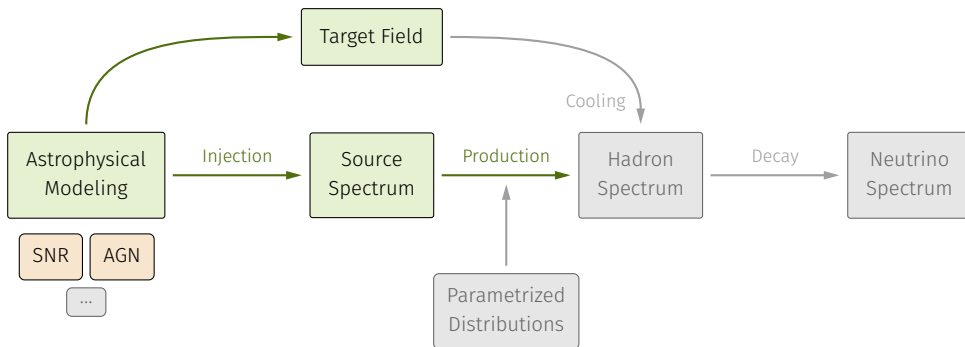
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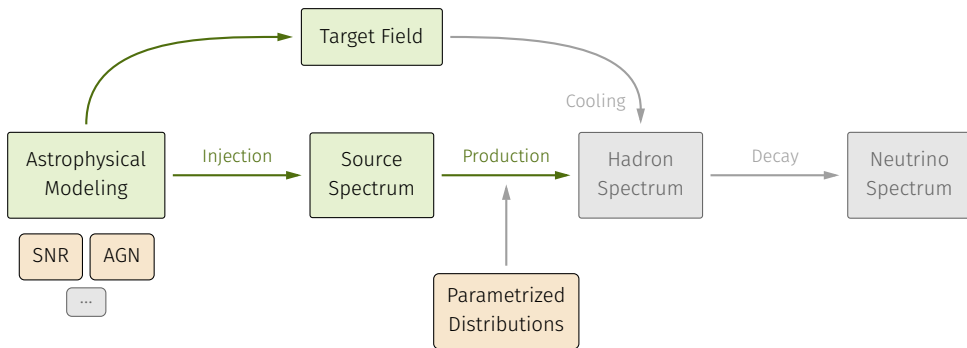
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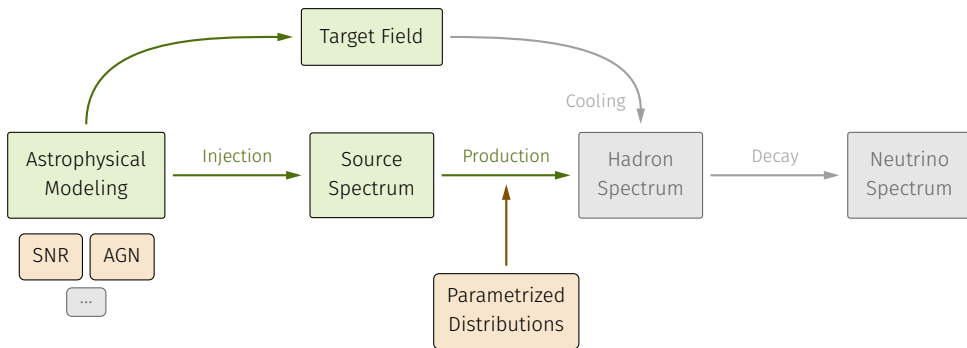
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$$\frac{dN_h}{dE_h} = \int_{E_h}^{\infty} dE_p \frac{dN_p}{dE_p} F_{pp \rightarrow h}$$









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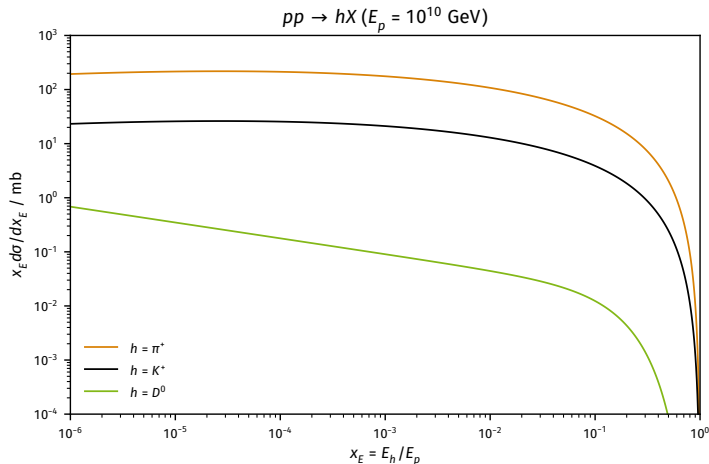
### ■ Energy Fractions

$$x_E = E_h/E_p$$

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$$z = E_h/E_c$$

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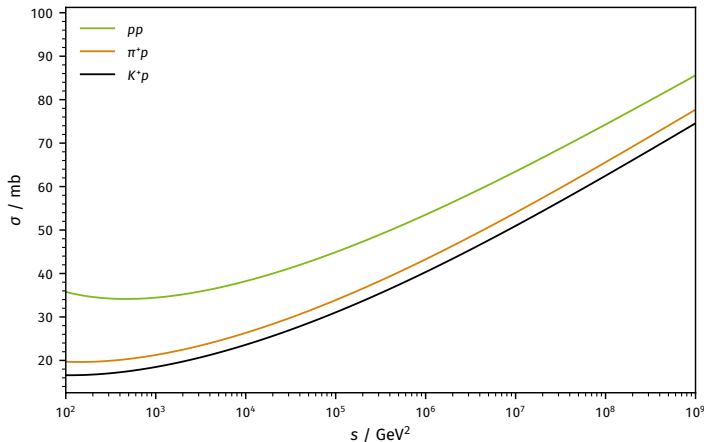


Derived Parametrizations

S. R. Kelner et al.  
*Phys. Rev. D* (2006)

V. P. Gonçalves et al.  
*JHEP* (2007)

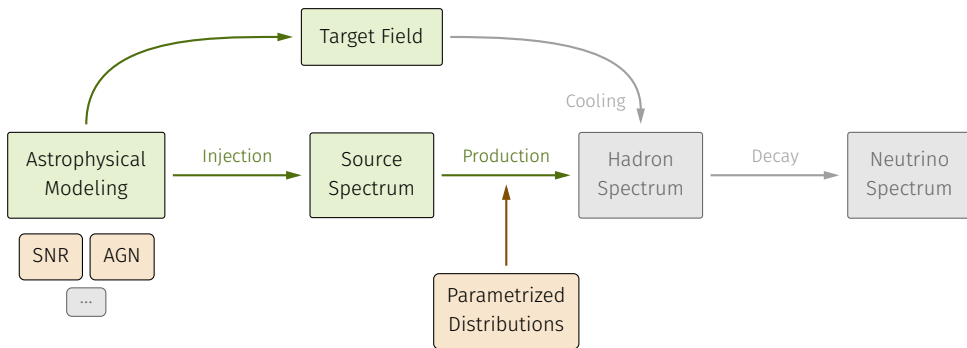
## Hadron Inelastic Scattering Cross Section

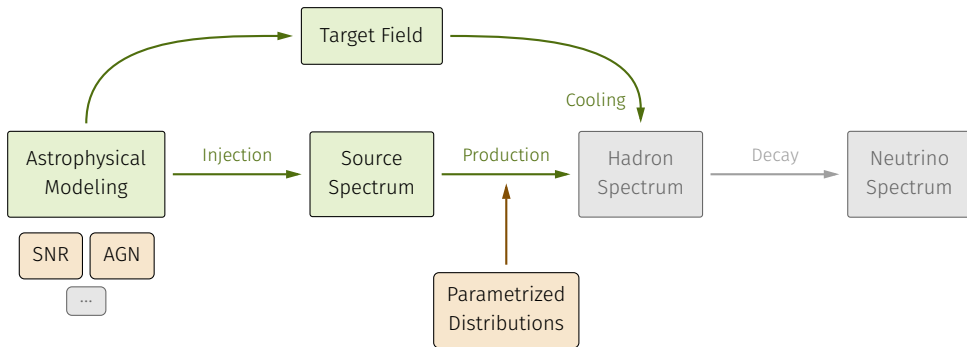


Combined Parametrizations

V. I. Belousov et al.  
*Phys. Atom. Nuclei* (2016)

D. A. Fagundes, M. J. Menon  
*Nucl. Phys. A* (2012)







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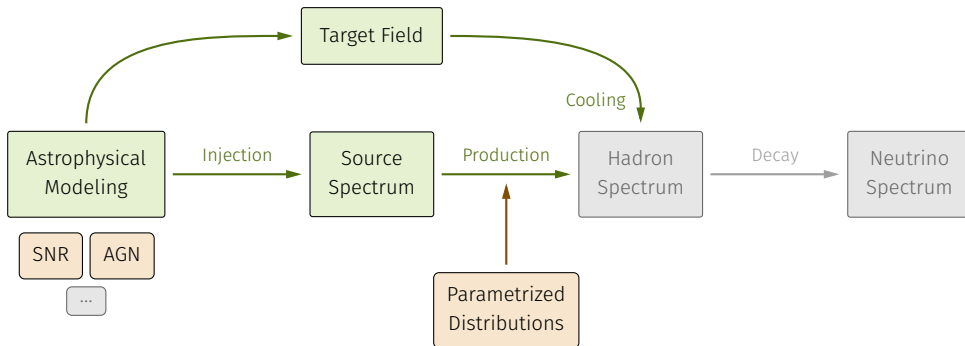
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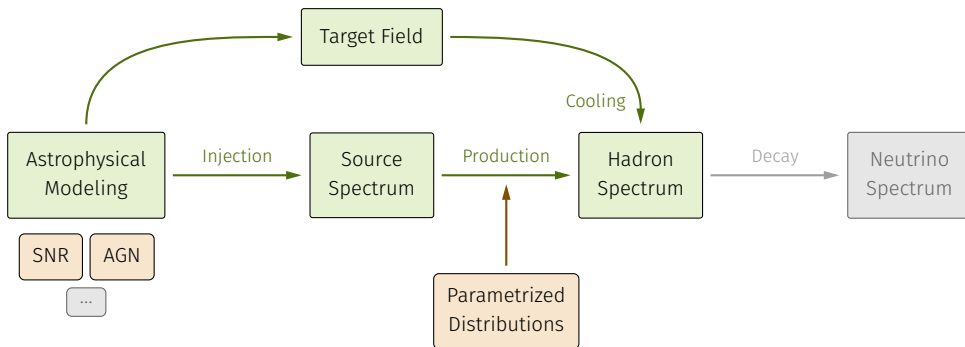
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$$O = \kappa_{pp} \sigma_{pp} n d$$

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| $D^+$         | $c\bar{d}$ | $1.0 \times 10^{-12}$ | 1.87               | 0.40                 | 0.176                   |
| $D_s^+$       | $c\bar{s}$ | $0.5 \times 10^{-12}$ | 1.97               | 0.71                 | 0.065                   |
| $\Lambda_c^+$ | $udc$      | $0.2 \times 10^{-12}$ | 2.29               | 1.61                 | 0.045                   |





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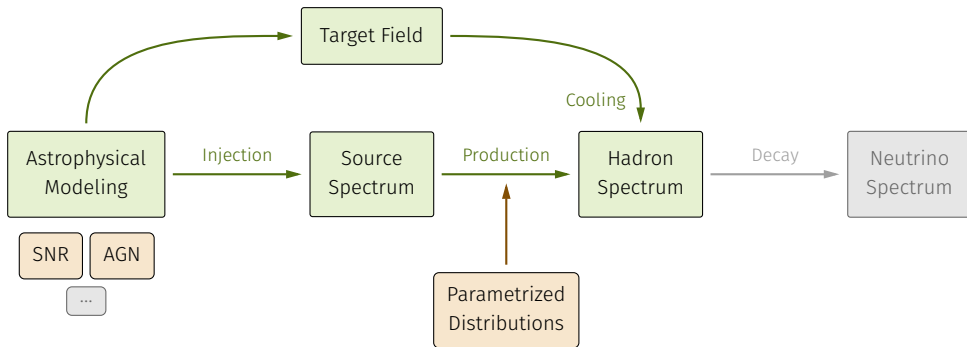
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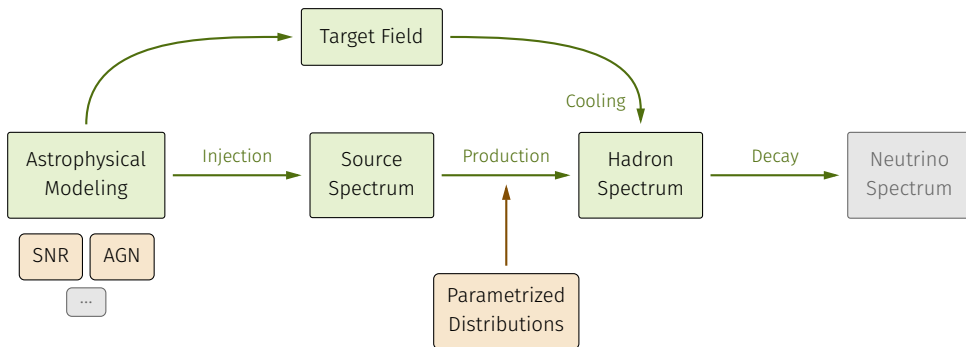
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$$\frac{dN_h}{dE_h} = CO \int_{E_h}^{\infty} dE_p \frac{dN_p}{dE_p} F_{pp \rightarrow h}$$

### ■ Temporal Evolution

$$CO \propto t^{-3} (1 - e^{-t^3})$$





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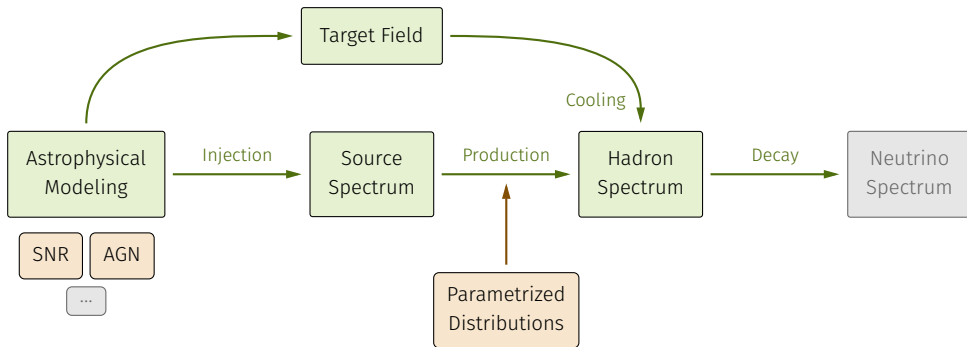
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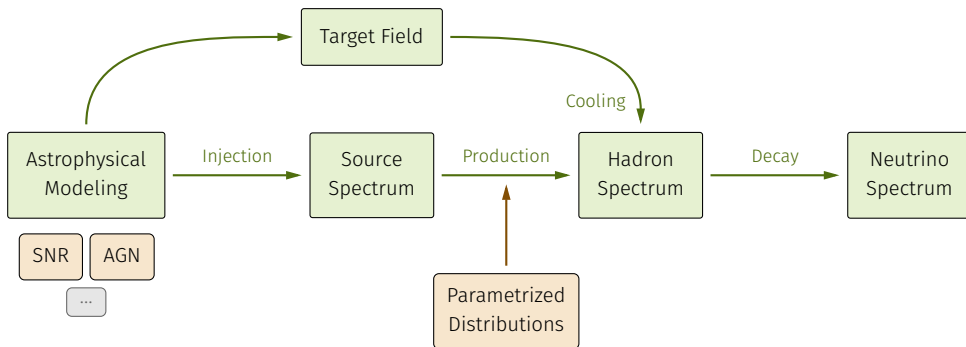
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$$\frac{dN_\nu}{dE_\nu} = \int_{E_\nu}^{\infty} dE_h \frac{dN_h}{dE_h} F_{h \rightarrow \nu}$$

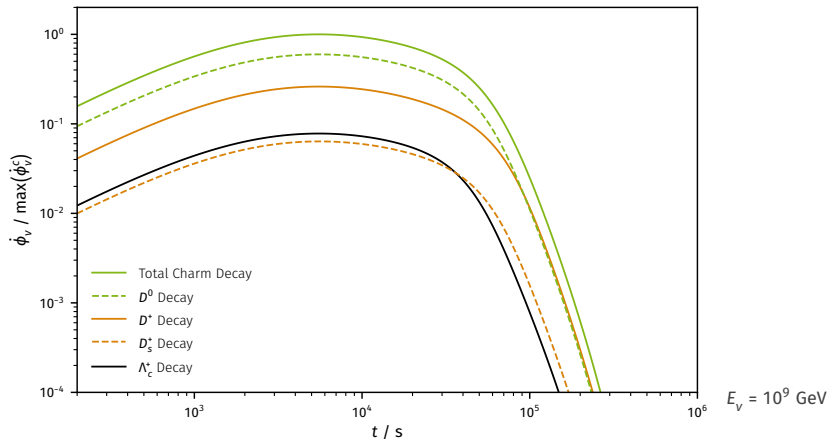
## Relevant Hadron Properties

| $h$           | $q_h$      | $\tau_h / \text{s}$   | $m_h / \text{GeV}$ | $s_h / \text{GeV}^2$ | $B_{h \rightarrow \nu}$ |
|---------------|------------|-----------------------|--------------------|----------------------|-------------------------|
| $p$           | $uud$      | $\infty$              | 0.94               | 15.98                | 0.000                   |
| $\pi^+$       | $u\bar{d}$ | $26.0 \times 10^{-9}$ | 0.14               | 10.23                | 1.000                   |
| $K^+$         | $u\bar{s}$ | $12.4 \times 10^{-9}$ | 0.49               | 12.62                | 0.636                   |
| $D_0$         | $c\bar{u}$ | $0.4 \times 10^{-12}$ | 1.86               | 0.45                 | 0.067                   |
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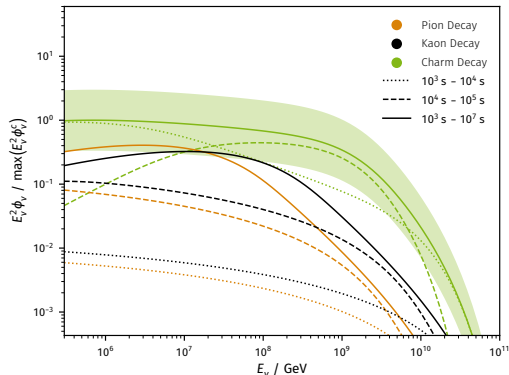
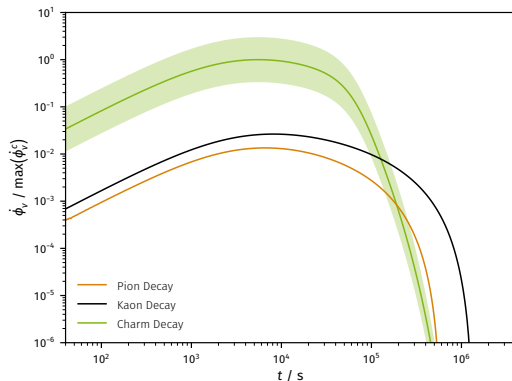




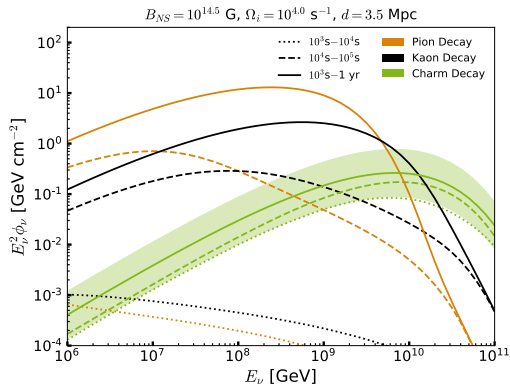
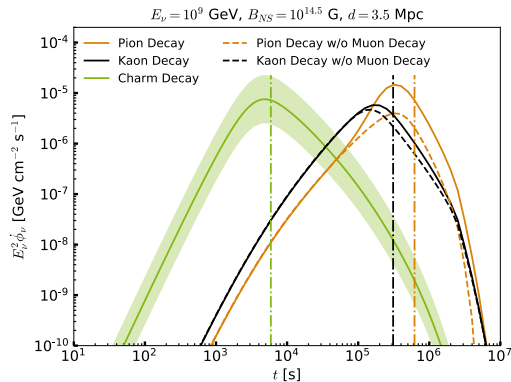
## Normalized Magnetar Charm Contributions



## Normalized Magnetar Neutrino Flux & Fluence

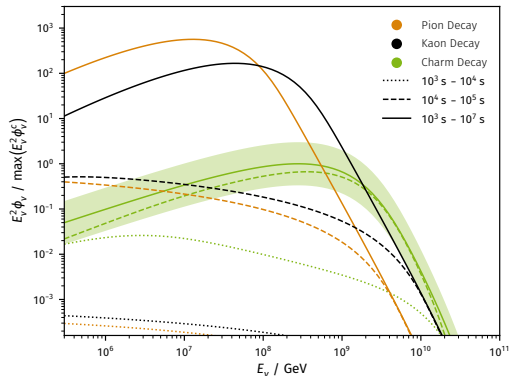
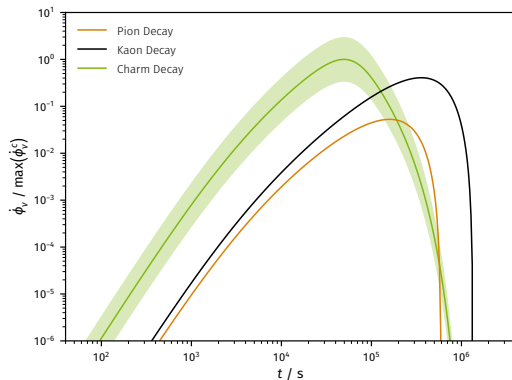


## Published Estimated Magnetar Neutrinos

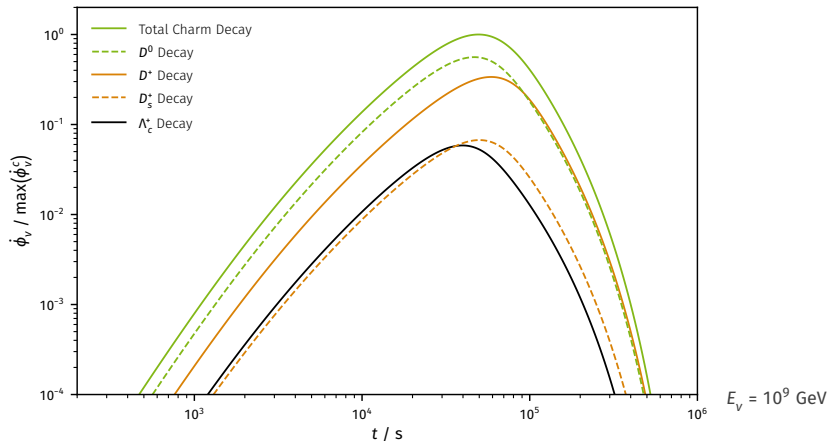


J. A. Carpio et al. *Phys. Rev. D* (2020)

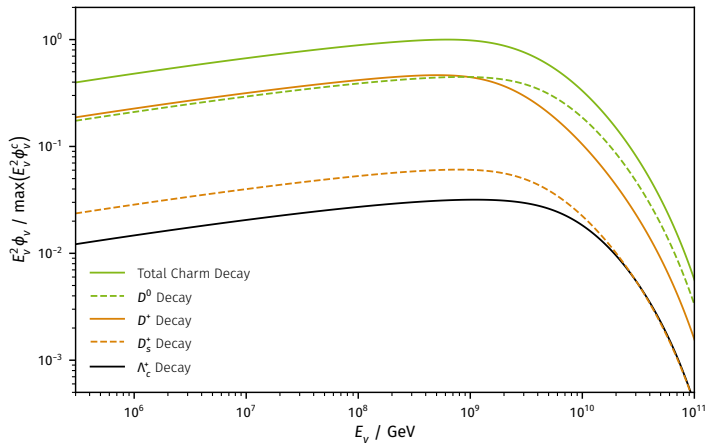
## Modified Magnetar Neutrino Flux & Fluence



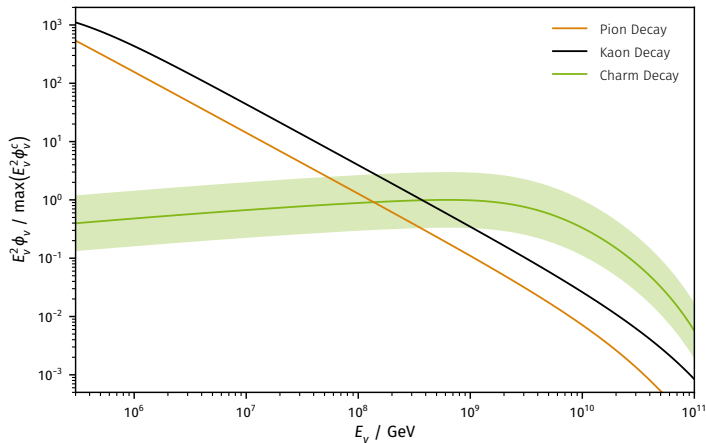
## Modified Magnetar Charm Contributions

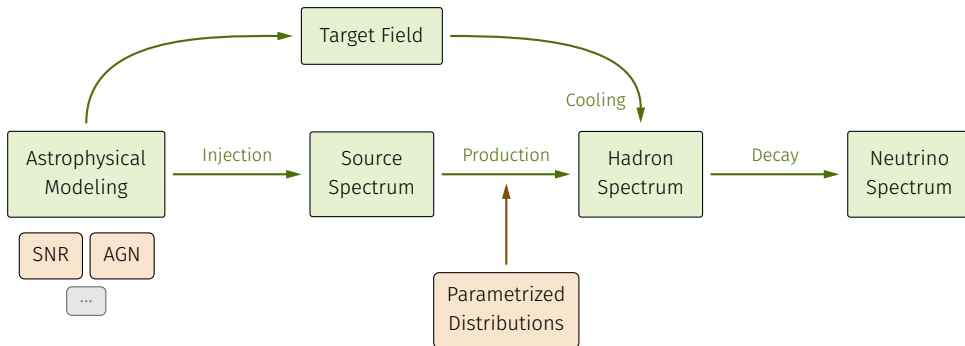


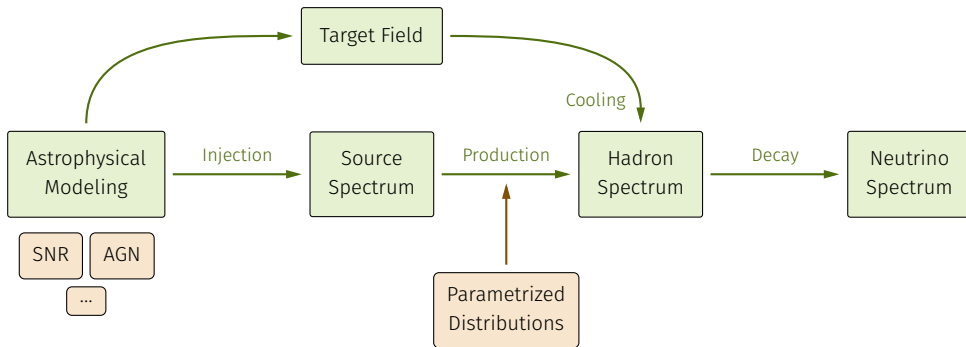
## Normalized AGN Charm Contributions



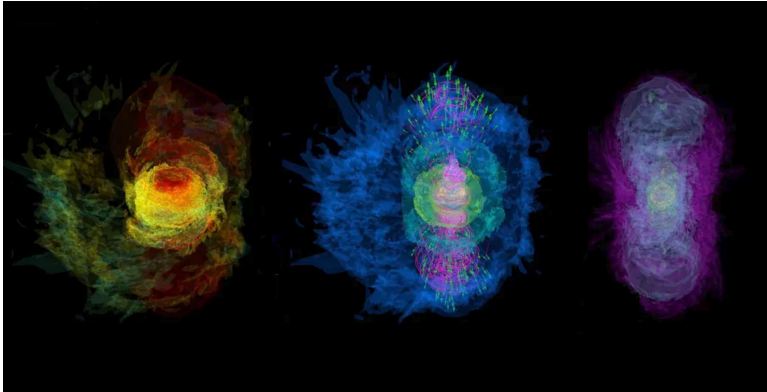
## Normalized AGN Neutrino Fluence





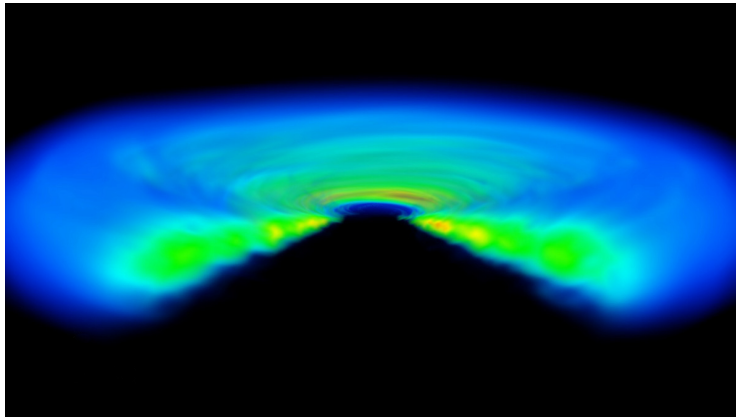


## Simulated Magnetar Formation

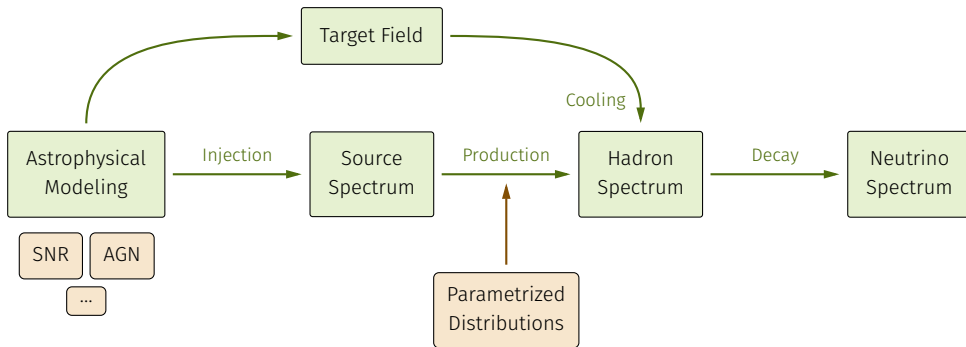


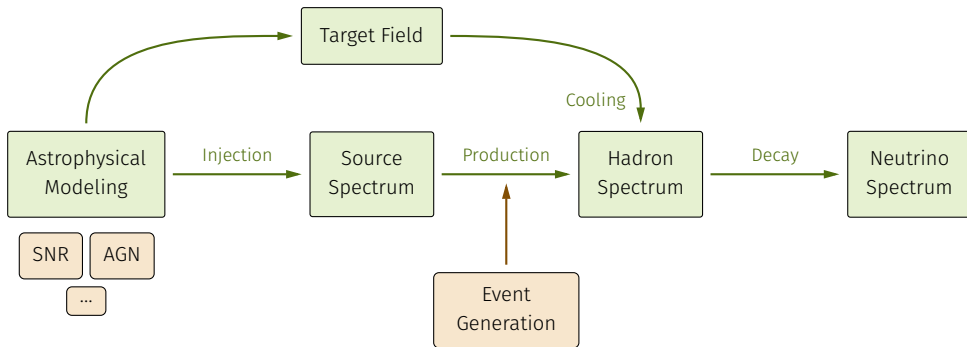
K. Hayashi *MPI Grav. Phys.* (2024)

## Accretion Disk Structure

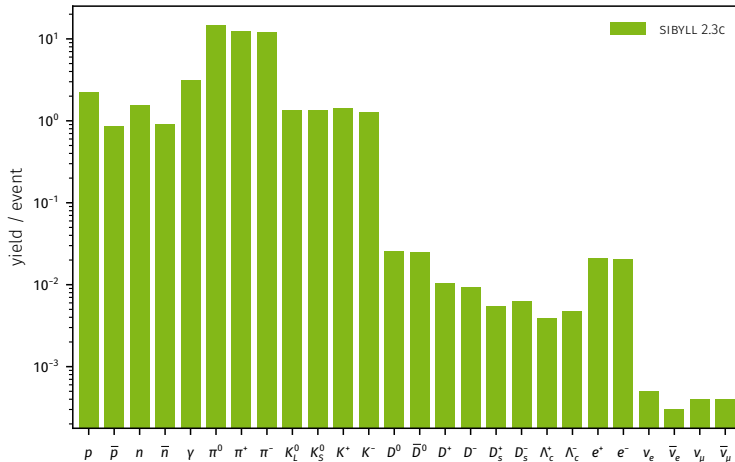


V. V. Zhuravlev et al. *ApJ* (2014)



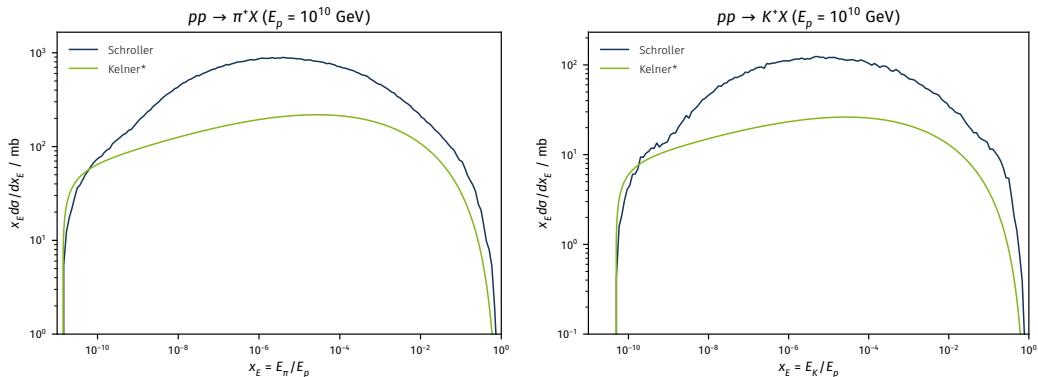


## Monte Carlo Event Generation



Data Courtesy  
M. Schroller

## Differential Cross Sections Comparison

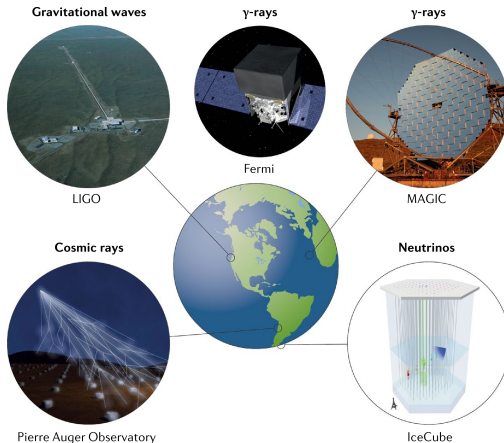


Derived Parametrization S. R. Kelner et al. *Phys. Rev. D* (2006)  
Event Data (SIBYLL 2.3c) Courtesy M. Schrollier



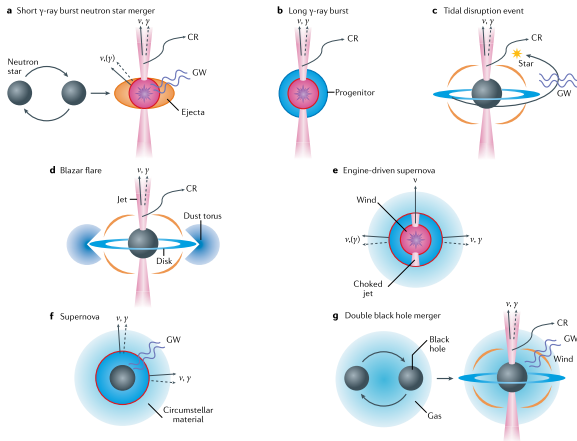
## Summary & Conclusion

## Messenger Types & Detection



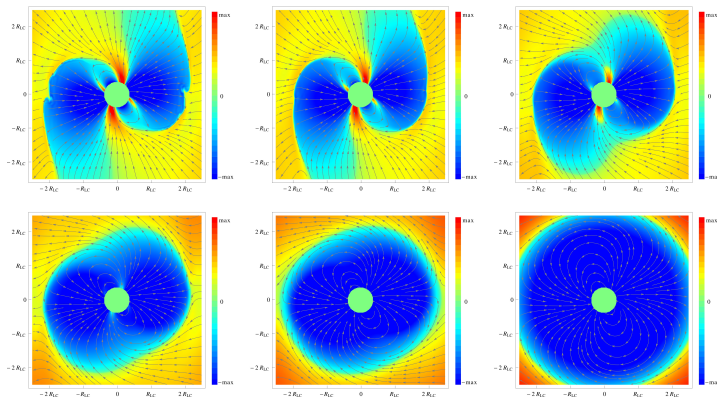
P. Mészáros et al.  
*Nat. Rev. Phys.* (2019)

## Source Regions & Events



P. Mészáros et al.  
*Nat. Rev. Phys.* (2019)

## Resistive Pulsar Magnetospheres



J. Li et al. *ApJ* (2012)

## Luminosity Models

### ■ Vacuum Dipole Radiation

$$L = \frac{2\mu^2\omega^4\sin^2\chi}{3c^2}$$

### ■ Force Free Magnetosphere

$$L = \frac{\mu^2\omega^4(1 + \sin^2\chi)}{c^3}$$

## Spindown Derivation

### ■ Magnetar Luminosity

$$L = K\omega^4 = -\dot{E}$$

### ■ Exponential Decay

$$E = E_0 \exp(-t/t_{sd}) \quad E = I\omega^2/2$$

$$\dot{E} = -E/t_{sd} \quad \dot{E} = I\omega\dot{\omega}$$

$$L_0 = K\omega_0^4 = I\omega_0^2/2t_{sd}$$

## Spindown Derivation

### ■ Spindown Time

$$t_{sd} = I/2K\omega_0^2$$

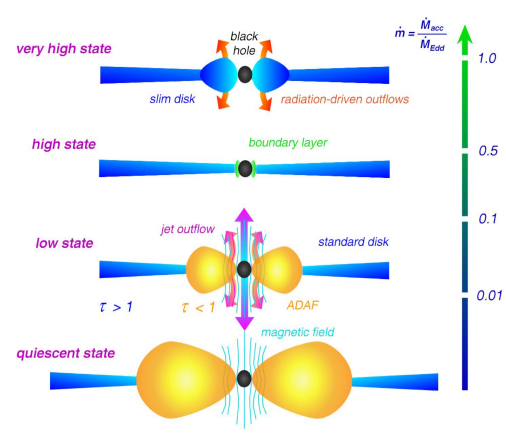
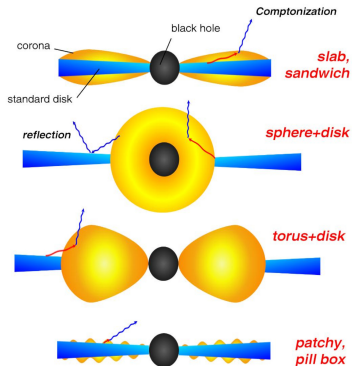
### ■ Differential Equation

$$I\dot{\omega}_0 = -K\omega^3$$

### ■ Separating Variables

$$\omega = \omega_0(1 + t/t_{sd})^{-1/2}$$

## Accretion Disk & Corona



A. Müller Heidelberg University (2004)